evolved emotional abilities and needs, and from the fact that pigs are mammals we can safely deduce that they have emotions.[16]

In recent decades life scientists have demonstrated that emotions are not some mysterious spiritual phenomenon that is useful just for writing poetry and composing symphonies. Rather, emotions are biochemical algorithms that are vital for the survival and reproduction of all mammals. What does this mean? Well, let's begin by explaining what an algorithm is. This is of great importance not only because this key concept will reappear in many of the following chapters, but also because the twenty-first century will be dominated by algorithms. 'Algorithm' is arguably the single most important concept in our world. If we want to understand our life and our future, we should make every effort to understand what an algorithm is, and how algorithms are connected with emotions.

An algorithm is a methodical set of steps that can be used to make calculations, resolve problems and reach decisions. An algorithm isn't a particular calculation, but the method followed when making the calculation. For example, if you want to calculate the average between two numbers, you can use a simple algorithm. The algorithm says: 'First step: add the two numbers together. Second step: divide the sum by two.' When you enter the numbers 4 and 8, you get 6. When you enter 117 and 231, you get 174.

A more complex example is a cooking recipe. An algorithm for preparing vegetable soup may tell us:

1. Heat half a cup of oil in a pot.
2. Finely chop four onions.
3. Fry the onion until golden.
4. Cut three potatoes into chunks and add to the pot.
5. Slice a cabbage into strips and add to the pot.

And so forth. You can follow the same algorithm dozens of times, each time using slightly different vegetables, and therefore getting a slightly different soup. But the algorithm remains the same.